

Principle of Communication System		
Lab #:	05	
Lab Title:	Amplitude Modulation (Full Carrier) Implementation using USRP	
Submitted by:		
Names		Registration #
Sumaira Safeer		FA22-BCE-019

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

LAB: 05

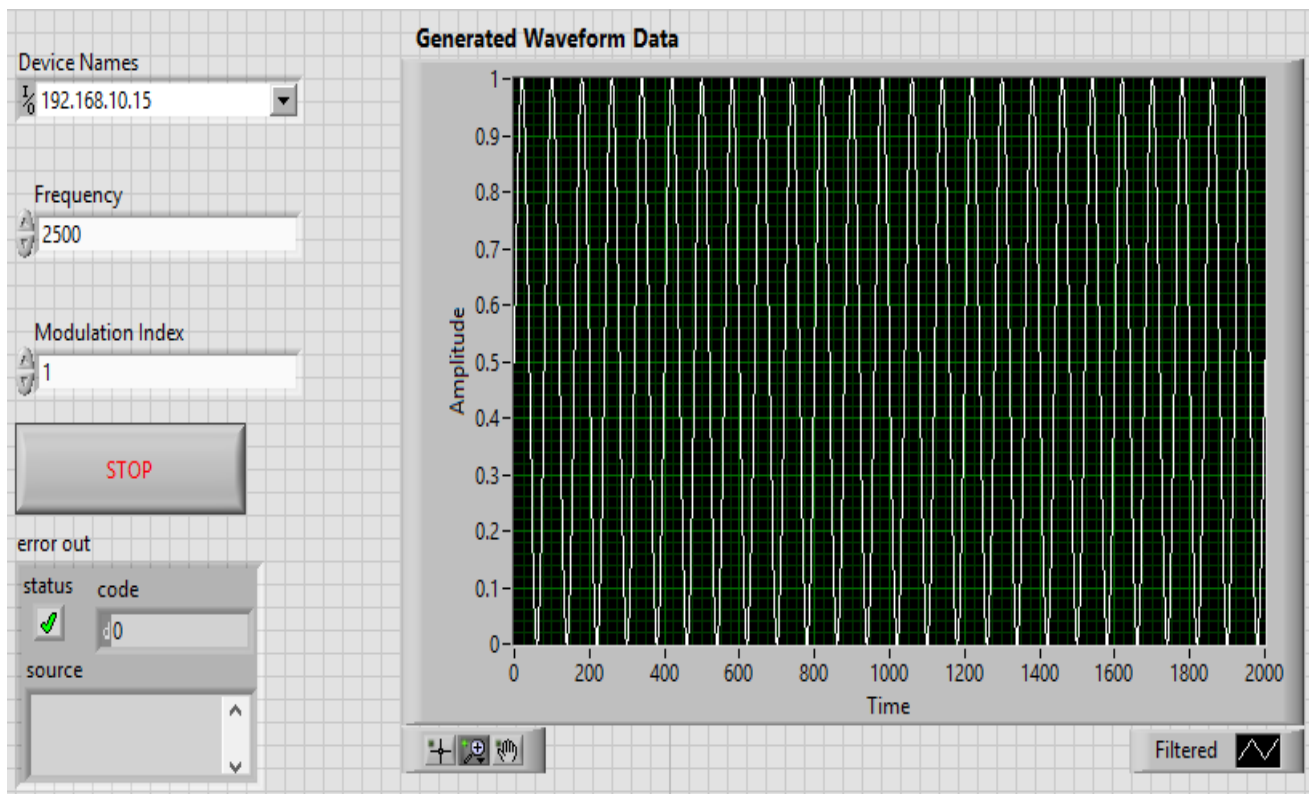
Topic: “Amplitude Modulation (Full Carrier) Implementation using USRP.”

Objective:

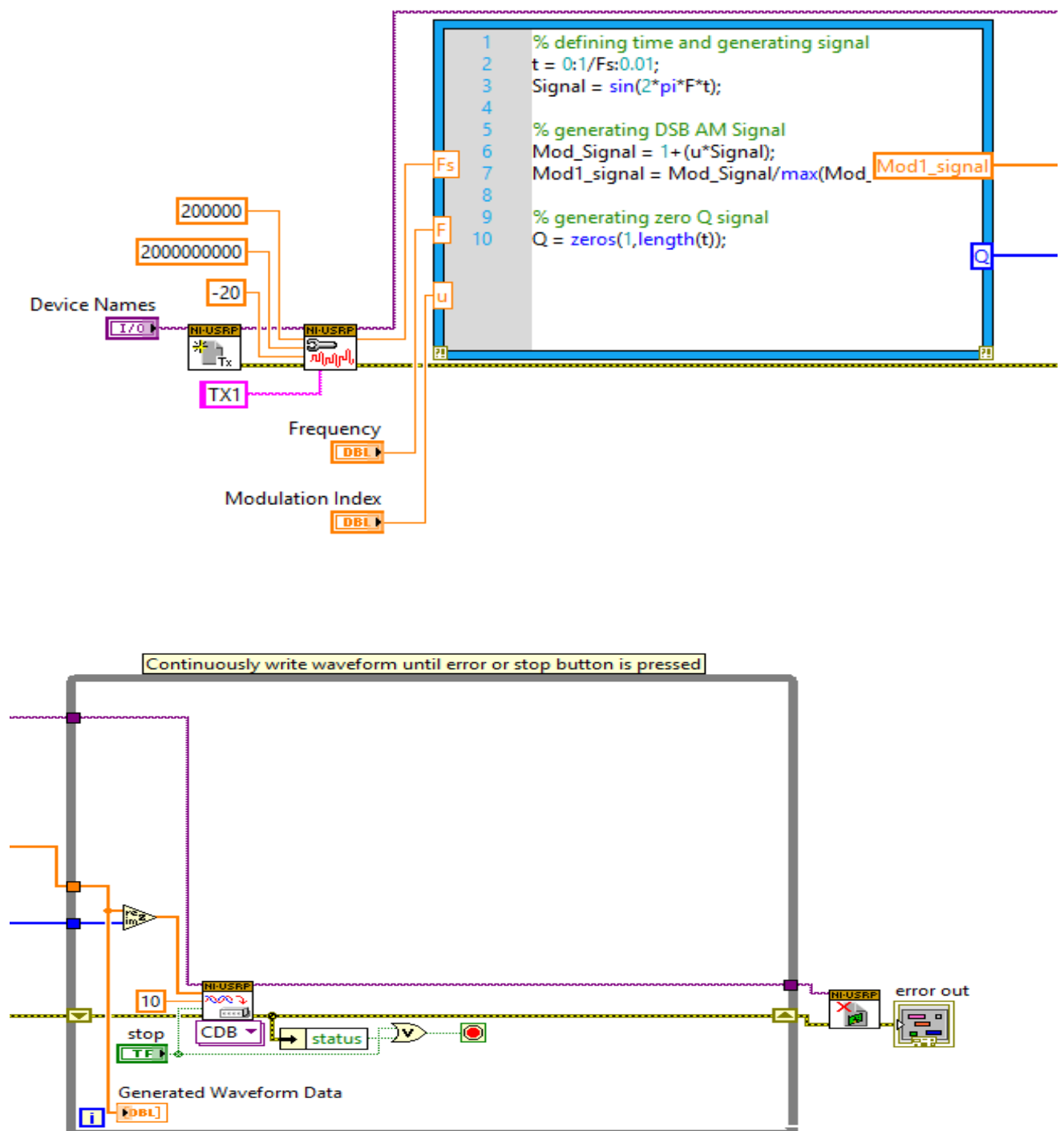
The main objective of this lab is AM implementation using USRP and LabVIEW.

AM Transmitter:

- FRONT PANEL:



Block Diagram:

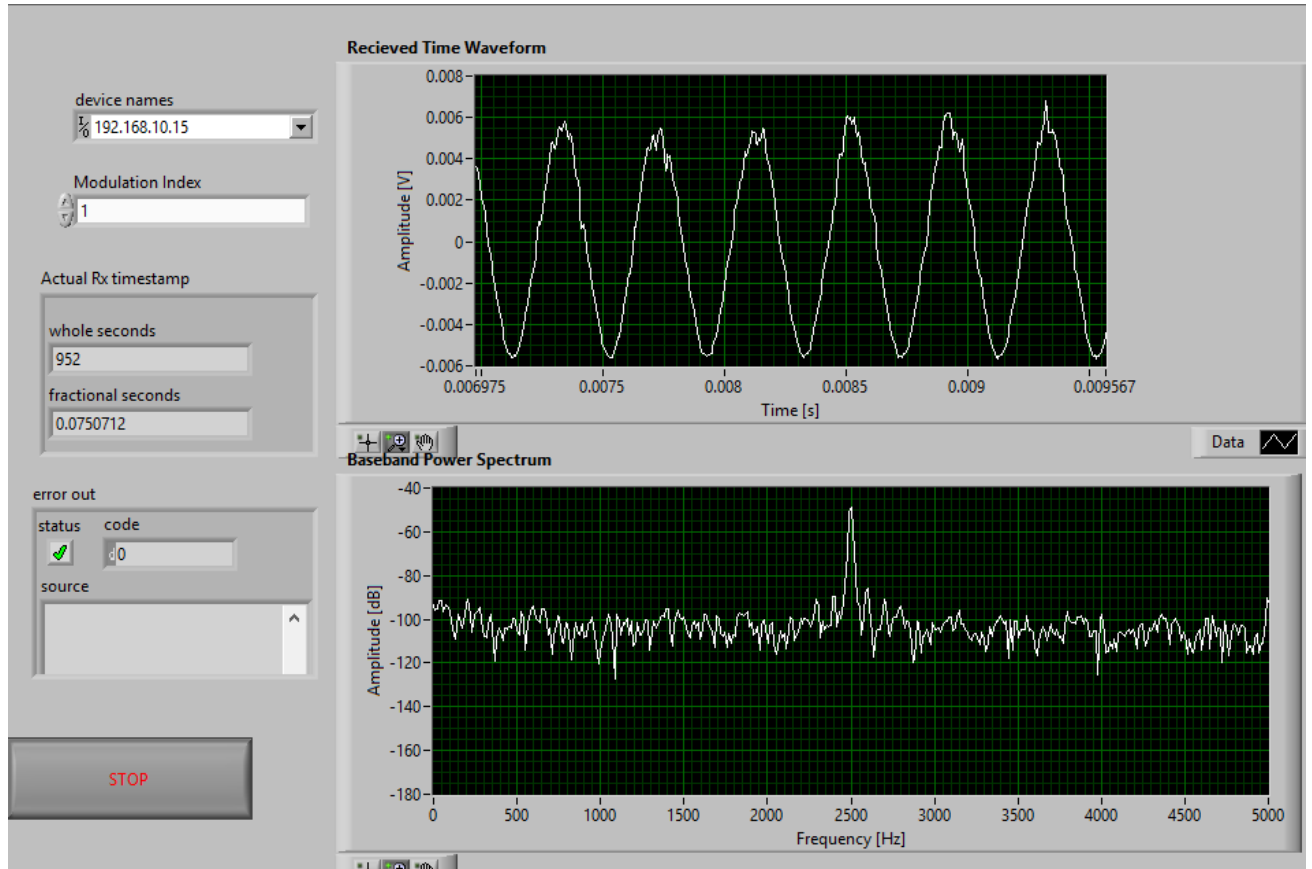


Conclusion:

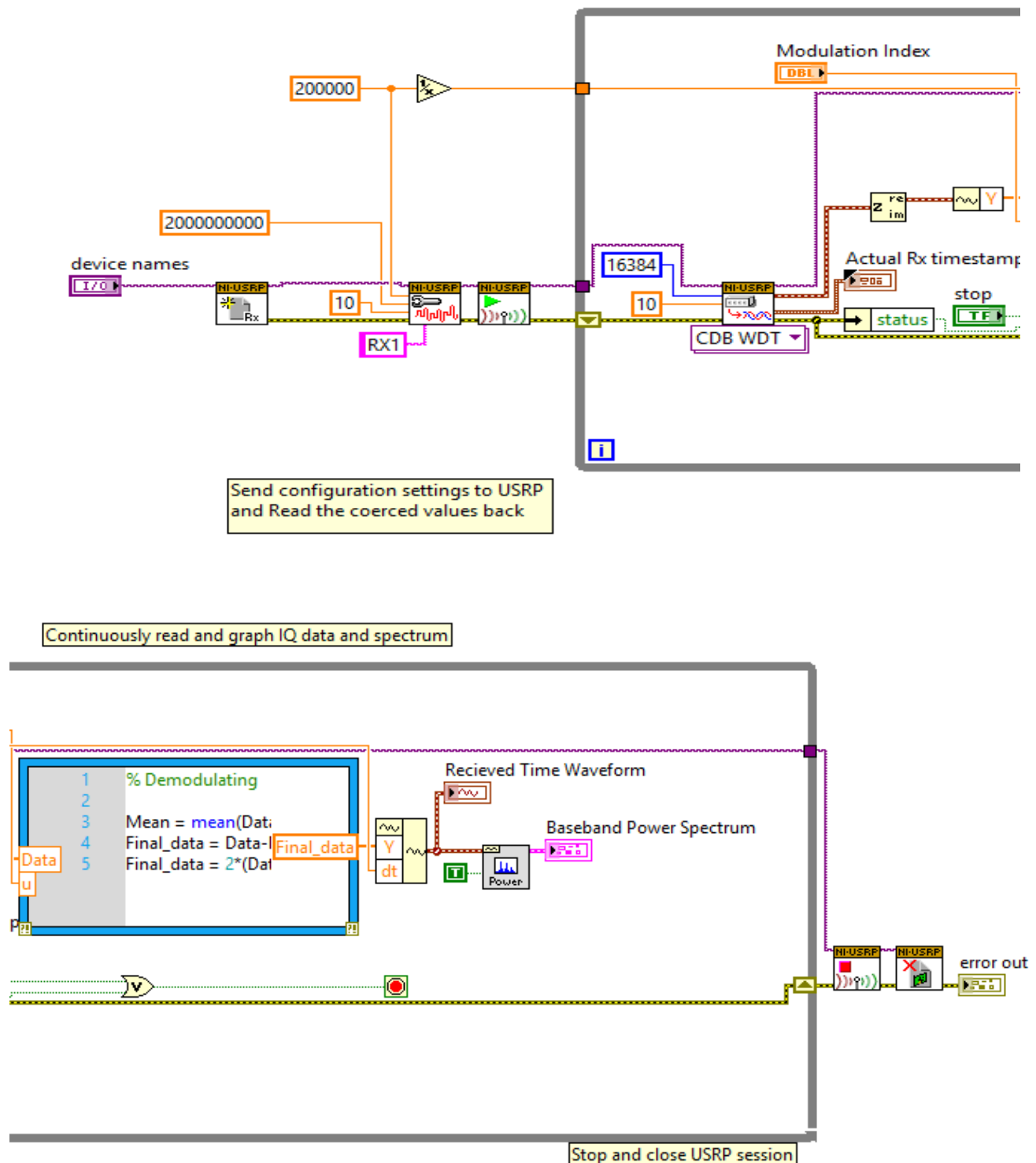
The AM transmitter successfully generates and transmits an amplitude-modulated signal using USRP and LabVIEW, effectively demonstrating the principles of amplitude modulation in a practical application. This implementation highlights the flexibility and capabilities of software-defined radio in real-time communication.

AM Receiver:

- FRONT PANEL:



Block Diagram:



Conclusion:

The AM receiver effectively captures and demodulates the transmitted signal, successfully reconstructing the original audio. This project illustrates the practical application of signal processing techniques and the reliability of USRP hardware in wireless communication systems.